

Classification of Legal Text: US Supreme Court Opinions by Issue Area

UE24CS352A Machine Learning mini-project · [Member 1 Name / SRN], [Member 2 Name / SRN] · github.com/AhanaSrinivas/legal-text-classification

1. Problem statement

Given the full text of a US Supreme Court opinion, predict its issue area: one of the 13 areas of the Supreme Court Database (SCDB), such as Criminal Procedure, Civil Rights or Economic Activity. It is single-label, multi-class classification of very long and strongly imbalanced documents. We reproduce the two methods of the reference project (Iyer, CS229 2020: LDA topics and Doc2Vec vectors, each with logistic regression), add TF-IDF linear baselines, and fine-tune Legal-BERT, which the reference project could not run because it ran out of memory.

2. Dataset details

- **Source.** LexGLUE SCOTUS ([coastalcph/lex_glue](https://github.com/coastalcph/lex_glue), config `scotus`; Chalkidis et al., 2022) with SCDB labels. 7,800 opinions in the official chronological splits: train 5,000 (1946–1982), validation 1,400 (1982–1991), test 1,400 (1991–2016). Columns: text, label.
- **Checks.** No empty texts; no exact duplicates within or across splits (SHA-256 of every text).
- **Imbalance.** Economic Activity has 1,043 training documents and Miscellaneous 3 (ratio 347.7:1). Rarest test classes: Miscellaneous (3), Interstate Relations (15), Attorneys (17).
- **Length.** Median 6,883 words per test opinion. On a 500-document sample the median is 5,470.5 Legal-BERT tokens, 93% of documents exceed 512 tokens, and the first 512 tokens cover about 20% of a document.

3. Approach

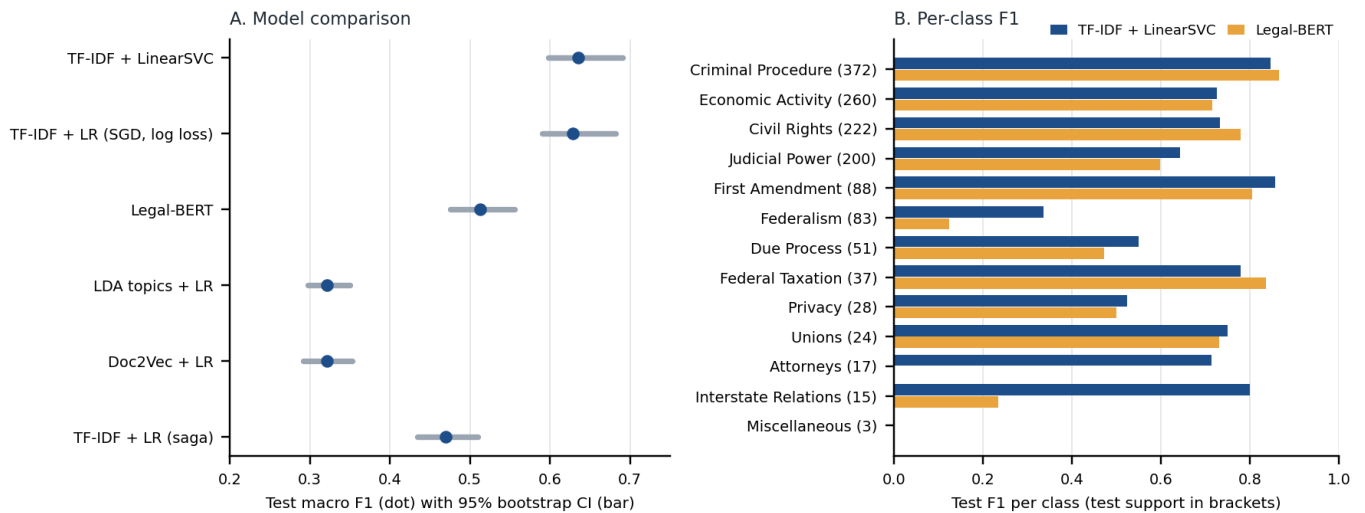
- **Protocol.** Hyperparameters chosen on validation only; final models fitted on train only; test predicted once per model. Primary metric: macro F1, so every issue area counts equally; we also report accuracy, weighted F1 and per-class scores with support, plus 95% percentile bootstrap CIs (1,000 resamples) for test macro F1.
- **Preprocessing.** NFKC normalisation, lower case, whitespace collapsed; standalone numbers removed (citations, docket numbers and years identify the era rather than the topic); scikit-learn English stop words minus words that change legal meaning (*not*, *no*, *shall*, *must*, *unless*, *without*, ...); no stemming. Vectorizers, LDA and Doc2Vec are fitted on the training split only.
- **Models.** (a) majority-class baseline; (b) TF-IDF (unigrams + bigrams, 10,000 features, sublinear tf) + LinearSVC (L2, squared hinge, $C = 1.0$ from 0.01, 0.1, 1.0); (c) the same features + logistic regression trained by SGD on log loss (L2, $\alpha = 1e-5$ from $1e-6$, $1e-5$, $1e-4$, converged after 9 of 50 epochs); an earlier saga run (max_iter 5, tol 0.5) is kept only as an under-converged reference; (d) LDA topic mixtures on raw counts + LR, $K = 40$ from 10, 20, 30, 40; (e) Doc2Vec PV-DM (100 dimensions) + LR, with vectors inferred for validation and test; (f) Legal-BERT (nlpueb/legal-bert-base-uncased) fine-tuned on the first 512 tokens: 3 epochs, effective batch 16, learning rate $2e-5$, weight decay 0.01, mixed precision, plain cross-entropy, best epoch by validation macro F1.

4. Brief implementation overview

Python with scikit-learn, gensim, PyTorch and Hugging Face transformers. `src/` holds reusable modules (data loading and audits, train-only preprocessing, shared evaluation); `scripts/` has one script per stage. Classical models were trained on a CPU laptop and Legal-BERT on a Colab Tesla T4 (19.6 minutes). Every model saves its validation and test predictions, and `evaluate_all.py` rebuilds all metrics, CIs, confusion matrices and the error analysis from them, so results reproduce without retraining. A Streamlit app (`app.py`) gives live predictions from the TF-IDF models and Legal-BERT next to the stored full-document predictions. The pytest suite checks preprocessing, leakage prevention and hand-computed metrics, and recomputes `metrics.csv` from the saved predictions. This write-up is generated from `results/` by `reports/writeup.py`.

Results

Model	Test macro F1	Test macro F1 95% CI	Test accuracy	Test weighted F1	Validation macro F1
TF-IDF + LinearSVC	0.635	[0.598, 0.691]	0.737	0.723	0.724
TF-IDF + LR (SGD, log loss)	0.629	[0.590, 0.682]	0.732	0.719	0.712
Legal-BERT (first 512 tokens)	0.512	[0.476, 0.556]	0.724	0.694	0.594
LDA topics + LR	0.321	[0.298, 0.350]	0.546	0.514	0.349
Doc2Vec + LR	0.322	[0.292, 0.353]	0.474	0.456	0.406
TF-IDF + LR (saga, under-converged)	0.470	[0.434, 0.510]	0.680	0.652	0.580
Majority class (train)	0.024	[0.022, 0.027]	0.186	0.058	0.021



Official LexGLUE test split; validation macro F1 is shown for reference only. CIs: 1,000 bootstrap resamples of the test set (seed 42). The saga row is the under-converged reference, not a final model. Per-class precision, recall, F1 and support for every model: `results/per_class_f1.csv`.

5. Conclusions

- Linear models on the full TF-IDF document are best: LinearSVC macro F1 0.635 (CI 0.598–0.691) and SGD logistic regression 0.629. Their intervals overlap, so we do not rank one above the other.
- Legal-BERT on the first 512 tokens has close accuracy (0.724 vs 0.737) but much lower macro F1 (0.512); its interval and LinearSVC's do not overlap, so the gap is not test-sampling noise. It never correctly predicts Attorneys and Miscellaneous and recalls only 7% of Federalism cases. Hypotheses, not tested: truncation to an opening that is mostly caption and citations, no class weighting, and a single short run (three identical runs ranged 0.5124–0.5319; we report 0.5124).
- The paper methods trail far behind (LDA 0.321, Doc2Vec 0.322), but in reduced CPU configurations, so this does not show the methods themselves are weak.
- The most frequent confusion is Federalism → Economic Activity for both LinearSVC (37 test cases) and Legal-BERT (49). The two models are partly complementary: 121 test cases are right only for LinearSVC and 103 only for Legal-BERT. Every trained model drops from validation to test (LinearSVC by 0.089), consistent with, but not proof of, drift across the chronological split.

Limitations

- Legal-BERT reads about 20% of a typical opinion (first 512 tokens).
- LDA (500 count features, 2 iterations; $K = 40$ at the grid edge) and Doc2Vec (5 epochs, first 500 tokens) ran in reduced CPU configurations.
- One seed per model; no class weighting. Rare classes have tiny test support (3, 15, 17 documents), so macro F1 is noisy.
- Bootstrap CIs are unpaired and capture test sampling only. The demo Legal-BERT weights come from the rerun, not the reported run.

References

- [1] K. Iyer. *Classification of Legal Text*. Stanford CS229 project report, Spring 2020.
- [2] I. Chalkidis, A. Jana, D. Hartung, M. Bommarito, I. Androustopoulos, D. M. Katz, N. Aletras. *LexGLUE: A Benchmark Dataset for Legal Language Understanding in English*. ACL 2022 (arXiv:2110.00976).
- [3] H. J. Spaeth, L. Epstein, J. A. Segal, A. D. Martin, T. J. Ruger, S. C. Benesh. *Supreme Court Database*, Version 2020 Release 01. Washington University Law.
- [4] I. Chalkidis, M. Fergadiotis, P. Malakasiotis, N. Aletras, I. Androustopoulos. *LEGAL-BERT: The Muppets straight out of Law School*. Findings of EMNLP 2020.